Department of Electrical & Electrical Engineering

BRAC University

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EEE383

Electronic System Design

Section: 04

Project Proposal

Smart Pedestrian Proximity Warning System

Group Number: 4

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Introduction

Crosswalks without traffic lights are dangerous zones where pedestrians frequently face road safety risks. While simple painted crosswalk lines are cheap, they fail to actively warn distracted drivers.

To solve this problem, our project introduces a low-cost standalone Dual-Sensor Smart Pedestrian Proximity Warning System. The system operates using two ultrasonic sensors that work together and are positioned right next to the crosswalk. Sensor 1 is responsible for the sidewalk and detecting pedestrians waiting for a chance to cross the road whereas Sensor 2 is responsible for detecting the approach of vehicles on the road. To save battery life, the system starts scanning vehicles and warning only if there are any pedestrians around the crosswalk.

Objective

Our goal is to build a compact circuit board (PCB) that integrates dual distance sensors, an ATmega328P microcontroller and warning alarms (bright LED lights and a loud buzzer). We will write software to clean up noisy sensor signals, measure vehicle distance in real time, and trigger distinct warning levels based on how close a vehicle gets to the crossing. Finally, we will test the complete prototype outdoors to make sure it works automatically and reliably in real-world weather conditions.

Main Mission

At its core our project aims to build a smart roadside guardian that watches the curb and measures the distance of coming cars using two ultrasonic sensors. Rather than running at full power non-stop, it follows a logical sequence to handle crosswalk traffic safely:

- It starts in standby mode. Sensor 1 constantly scans the sidewalk area. If no one is standing at the curb, the rest of the unit stays inactive so cars driving by an empty crosswalk won't set off false alarms.
- It wakes up when a pedestrian arrives. The moment someone steps up to cross, Sensor 1 spots them and immediately signals Sensor 2 to start tracking oncoming cars down the lane.
- It raises a caution alert if a car approaches. If a vehicle enters the outer distance threshold, the system triggers a slow 2 Hz light flash and intermittent beeps to give both the driver and pedestrian an early heads-up.
- It switches to a full danger alarm if the car gets too close. If the vehicle crosses the inner safety limit, the LEDs instantly shift to a fast 6 Hz strobe while the buzzer sounds a loud, continuous 85 dB siren.

To make the device work well on different streets, we also added two simple dial knobs (potentiometers) directly onto the board letting us manually fine-tune the exact caution and danger distance boundaries whenever needed.

System Specifications & Technical Requirements

Sl.	System Requirement	Target Specification
1	Detect waiting pedestrians at the crosswalk curb (Sensor 1)	0.3 to 1.5 meters
2	Detect approaching vehicles down the traffic lane (Sensor 2)	0.3 to 4.0 meters.
3	Allow users to adjust the caution and danger distance limits	0.5 to 3.5 meters via potentiometers
4	Trigger the alarm quickly once a limit is breached	Under 200 milliseconds total latency
5	Flash the LED lights at different speeds based on the danger level	2 Hz for caution, 6 Hz for danger
6	Sound a clear warning buzzer depending on the alert level	At least 85 dB from 10 cm away
7	Filter out bad sensor data to prevent false alarms	3-sample median filter zero false alarms in a 1-hour idle test
8	Run safely from a standard DC power source with regulated voltages	9–12V input with surge and reverse-polarity protection
9	Operate reliably outside in various weather and lighting conditions	0–50 °C with an IP54-rated weatherproof enclosure
10	Keep the power and signal areas separate on a small circuit board	2-layer PCB, maximum 80 × 60 mm

Name of Subsystems

- Power Subsystem (Regulation and Protection)

This part will accept the standard 12V DC input and step it down to a regulated 5V to power all the logic and analog components. The diodes and filtering capacitors integrated in this sub system will work as a protection towards reverse polarity in case the plug is ever connected backwards as well as provide voltage surge protection.

- Processing and Control Subsystem

An ATmega328P microcontroller will be working as the core of the whole system, and coordinate dual-channel timer bursts, execute 3-sample median filtering to remove noise, and evaluate real-time distance thresholds. Or simply, prevent interference between the sensors, help prevent false detection and manage the trigger condition.

(The system will filter out environmental noise using the LM358 op-amp to filter out random noise picked along with the 40kHz signals, followed by the ATmega328P software handling the median filtering to clean up reading spikes before deciding to trigger the warning.)

- Sensing Subsystem

By utilizing two discrete 40 kHz ultrasonic transceiver pairs (TCT40-16T/R) with dedicated dual LM358 operational amplifiers, the system can capture the required data from the echoes. One sensor will monitor the curb/sidewalk/footpath to check if there are any pedestrians to aid in crossing; this works like a wake-up feature to make the system more efficient and prevent unnecessary noise or detection when it is not needed. Another sensor will scan for the oncoming traffic (while filtering out environmental noise) which the system will warn the pedestrians of. Both the readings will be sent to the control unit as data to process.

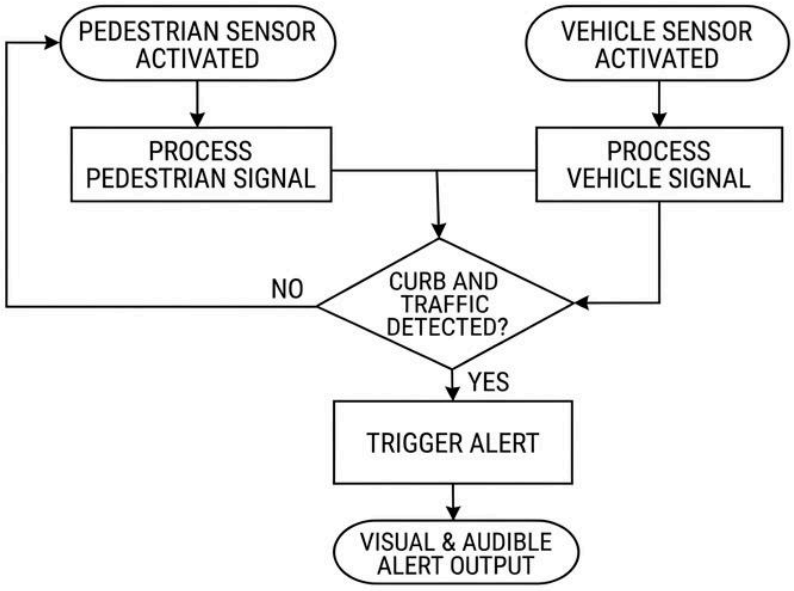
- UI and adjustment Subsystem

This part serves the purpose of adjusting the warning zones in both directions, for both the pedestrian and oncoming traffic, with the use of 10k potentiometers paired with RC low-pass filters.

- Actuators and Power Driver Subsystem

The microcontroller cannot supply the output with required voltages. Instead, it will work with low-side IRFZ44N N-channel MOSFET drivers, switching the high current output devices including high-brightness 1W SMD LED and an 85 dB piezo electronic alarm siren.

Methodology



The operational methodology follows a discrete hardware processing pipeline divided into four main stages:

Dual Ultrasonic Sensing & Signal Conditioning (TCT40-16T/R & LM358)

- Transducer Transmit: ATmega328P hardware Timer1 will generate 40 kHz square wave bursts through two TCT40-16T transmitters:
    - Sensor 1 (Pedestrian): Aimed at the sidewalk waiting area (0.3–1.5 m).
    - Sensor 2 (Vehicle): Aimed down the oncoming traffic lane (0.3–4.0 m).
- LM358 Analog Conditioning: Echo signals received by the TCT40-16R transducers are very small amplitude AC waves. Each channel is regulated through two exclusive LM358 dual OP Amp ICs:
    - Stage 1 (Bandpass Amplifier): It will eliminate the low frequency noise from the environment and only amplify the 40kHz signal.

    - Stage 2 (Peak Detector & Comparator): It will transform the reflected waveform into a noiseless 5V signal which will be sent to the microcontroller.
- Time of Flight (ToF): We will calculate the distance, $d = (v \times t) / 2$, where $v \approx 343$ m/s (speed of sound in air).

Digital Noise Filtering & Two Condition Alert Logic

- 3 Sample - Median Filter: The microcontroller takes 3 short distance readings on both channels and picks the center value which eliminates abnormal readings(leaves, birds, etc.)
- Two State Decision Machine:
    - No pedestrian ($d_{ped} > 1.5$ m): System will remain in idle/standby mode. Passing traffic does not trigger an alarm.
    - Pedestrian present($d_{ped} \leq 1.5$ m):Will actively monitor vehicles:
        - Vehicle Far ($d_{vehicle} > D_{caution}$): The LEDs will have constant low flash indicating the system is on.
        - Caution ($D_{danger} < d_{vehicle} \leq D_{caution}$): The LEDs will flash at 2 Hz and the buzzer will beep intermittently at 2 Hz.
        - Danger ($d_{vehicle} \leq D_{danger}$): The LEDs will strobe at 6 Hz and the buzzer will sound continuous 85 dB siren.

Distance Threshold Tuning & MOSFET Drivers

- Potentiometer Adjustments: Using two 10k potentiometers connected to ADC pins we can adjust Caution ($D_{caution}$) and Danger (D_{danger}) boundaries (0.5–3.5 m).

- MOSFET Driving: IRFZ44N MOSFETs will be used for switching for 12V high-power loads, including 1W LEDs and an 85 dB piezo alarm.

Power Supply & Circuit Protection

- Circuit Protection & Regulation: We will use an 1N4007 diode to prevent reverse polarity. And to clamp the voltage waveform we will use an 1N4744A Zener diode. An AMS1117-5.0 linear regulator will step down 12V DC to a stable 5V rail.

Components of All Subsystems

Component Name	Purpose	Estimated cost
Subsystem : Power		
DC Barrel Jack (12V DC Adapter)	Power Input	40.00
1N4007 diode	Reverse Polarity Protection	4.00
1N4744A (in parallel) (pair with regulator AMS1117)	Surge Protection	5.00
AMS1117-5.0 (LM7805 during prototyping)	Voltage Regulation	50.00
Filtering Capacitors (2x 10µF Electrolytic, 2x 0.1µF Ceramic)	Input/output bulk filtering and MCU/regulator noise decoupling	15.00
Subsystem : Processing and Control		
ATmega328P-PU	Microcontroller	332.00
16 MHz Quartz Crystal oscillator	Clock	18.00
10K Resistor + Push Button	Reset	5.00 + 5.00
AVR ISP 10 Pin to 6 Pin Adapter Board	Microcontroller connection	95.00
Subsystem : Sensing		
40KHZ 16mm Ultrasonic Sensor 40KHZ Receiver Transmitter R and T TCT40-16T/R **(2x)**	Distance Sensor	198.00
LM358 Dual Operational Amplifiers Op-Amp DIP8 (pair with appropriate resistors and capacitors to configure the op amp) **(2x)**	Amplify and Filter signals	64.00

Subsystem : UI & Adjustments		
2x 10k Potentiometers	Adjusting warning distance	20.05
2x 0.1uF Ceramic Capacitors		5.00
Subsystem : Actuators & Power Driver		
2x IRFZ44N MOSFET	Switch the outputs on/off	110.00
1W SMD LED Light Beads	Lights	40.00
Piezo Electronic Buzzer	Alarm	95.00
150 to 220 ohm Resistors		30.00

Components such as resistors and capacitors are subject to change upon further development.

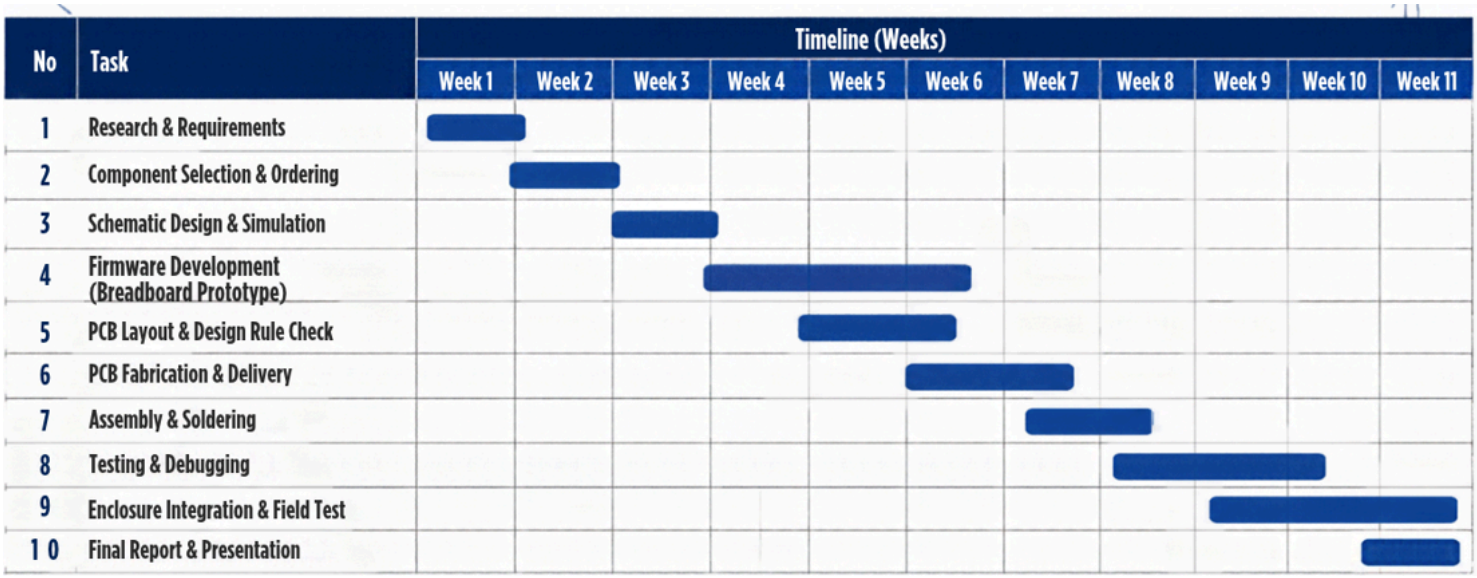
Budget Analysis

No.	Component Description	Quantity	Unit Price (BDT)	Total Cost (BDT)
1	Microcontroller (ATmega328P-PU)	1	332.00	332.00
2	Ultrasonic Sensor (TCT40-16T/R Pair)	2	99.00	198.00
3	Power Driver (IRFZ44N MOSFET)	2	55.00	100.00
4	High-Brightness LEDs (1W SMD Beads)	1	40.00	40.00
5	Alarm (Piezo Electronic Buzzer)	1	95.00	95.00
6	Voltage Regulator & Protection (AMS1117-5.0, Diodes)	1 Set	59.00	59.00

7	Signal Conditioning (LM358 Dual Op-Amp)	2	32.00	64.00
8	Threshold Tuning (10k Potentiometers & Caps)	2 Sets	25.05	25.05
9	System Hardware (Power Jack, Crystal, Switches, Adapter)	1 Set	163.00	163.00
10	Custom PCB Fabrication & Enclosure	1 Set	2,400.00	2,400.00
	Subtotal			3,461.05
	Contingency (10%)			346.11
	Estimated Total			3,807.16 BDT

Project Timeline (Gantt Chart)

No.	Task	Start	End	Duration
1	Research & Requirements	Week 1	Week 2	2 Weeks
2	Component Selection & Ordering	Week 2	Week 3	2 Weeks
3	Schematic Design & Simulation	Week 3	Week 4	2 Weeks
4	Firmware Development	Week 4	Week 6	3 Weeks
5	PCB Layout & Design Rule Check	Week 5	Week 6	2 Weeks
6	PCB Fabrication & Delivery	Week 6	Week 8	2 Weeks
7	Assembly & Soldering	Week 7	Week 8	2 Weeks
8	Testing & Debugging	Week 8	Week 10	3 Weeks
9	Enclosure Integration & Field Test	Week 9	Week 11	3 Weeks
10	Final Report & Presentation	Week 10	Week 11	2 Weeks



Conclusion

Safe pedestrian crossing is absolutely necessary and this is exactly what our project does. It is a highly reliable and economical solution to safeguarding pedestrians without the false alarms for motorists. With two-way ultrasonic sensor technology, customized op-amp circuits, fast data processing, and audio-visual indications, our design exhibits considerable engineering discipline. With a budget of 3,807.16 BDT and a well-defined 12-week schedule we are all set to complete the circuit design and prototype building process.